

Date: 07 July 2026	Team ID: SB-7429
Project Name: Smart Lender: Applicant Credibility Prediction for Loan Approval	Maximum Marks: 2 Marks

TECHNOLOGY STACK TEMPLATE

Define Your Technology Stack

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	Frontend User Interface	HTML5, CSS3, JavaScript (ES6)	Allows building a responsive dark-themed fintech dashboard with smooth transitions, custom tabs, and grid layouts without third-party frameworks.
2	Backend Web Server	Python, Flask	Lightweight, Python-native micro-framework. Perfect for loading model pickles, standardizing form payloads, and rendering templates.
3	Machine Learning Pipeline	Pandas, NumPy, Scikit-Learn, XGBoost	Robust data manipulation (Pandas/NumPy) and machine learning benchmarking, with XGBoost selected for high ensemble predictive accuracy (83.7% validation accuracy).
4	Serialization Layer	Python Pickle (.pkl files)	Fast, lightweight binary serialization format to save standard scaler weights, feature names list, and trained model weights.
5	Analytics Charts	Matplotlib, Seaborn	Generates high-resolution EDA count plots and scatter charts during the training phase, saved directly into static assets.